



SPRING MID SEMESTER EXAMINATION-2024

School of Computer Engineering
Kalinga Institute of Industrial Technology, Deemed to be University
Computer Organization and Architecture
[CS21002/ CS 2006]

Time: 1 1/2 Hours

Full Mark: 20

*Answer Any four questions including question No.1 which is compulsory.
The figures in the margin indicate full marks. Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.*

1. Answer all the questions. [1 Mark X 5]

- a) Consider the processor of a computer which can support a maximum memory of 16TB. Assume that memory is word addressable and a word consists of 8 bytes. The size of address bus of the processor is at least 41 bits.
- b) Explain the basic steps required to perform any instruction using 5 special purpose registers.
- c) How does Harvard architecture differ from Von Neumann architecture? Why Harvard architecture is not practically used?
- d) A word addressable computer has 8GB of memory and the word size is 64 bits. Assuming there are 80 control lines and 40 interrupt lines, how large is the system bus?

How to represent a number "-65" in little and big endian style. Assuming the memory locations start at the address 2000 and the memory is byte addressable organized in 16-bit word.

Big: FF BF 65 = 0000 0000 0100 0001
Little: BF FF -65 = 1111 1111 1011 1101
 F F B F

2. i.) Explain the steps of execution of the following instructions with respect to the use of registers in the processor. [2 Marks]

Address	Instruction
1000	Load LOCA, R1
1004	ADD R1, R0

(Results stored in last operand in instruction)

ii.) A two word instruction is stored at location 1070 with its address field at location 1071. The address field has the value 1205 and the value stored at 1205 is 2803 and at 2803 is 1200. The words stored at 1204 and 1206 are 1500 and 4800 respectively. A processor register R1 has value 1205. Evaluate the effective address and operand if addressing mode of the instruction is as follows : [3 Marks]

(a) Auto Increment (b) Auto Decrement

EA = 1205

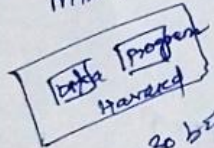
operand = 2803

EA = 1204

operand = 1500

$$\frac{44}{2} = 22$$

IR, PC, MAR,
MDR, SP



Addr. bus = 30 bits
Data bus = 64 bits
Control = 120 bits
Total = 214 lines

3.

i.) Consider the code given below and answer the following questions. All the instructions are 4 bytes in length. [3 Marks]

(a.) What will be the offset for the instruction BGTZ (Branch greater than zero) to branch to the location labelled LOOP?

(b.) What will be the content of PC when R1 becomes zero?

offset = -12
(1024 - 1024)

PC = 1024,
R1 = 0

1012

1000	MOV #10, R1
1004	MOV #LOCA, R2
1008	CLEAR R3
1012	LOOP ADD (R2)+, R3
1016	DEC R1
1020	BGTZ LOOP
1024	CALL SUB1

ii.) Describe briefly about the 3-address and 2-address instruction format. Write a program to evaluate the following arithmetic expression [2 Marks]

$$X = (A + B * C) / (D - E * F + G * H)$$

using general purpose register with

a) 3-address and b) 2-address instruction format.

4. Consider two different machines, with two different instruction sets, both of which have a clock rate of 200 MHz. The following measurements are recorded on the two machines running a given set of benchmark programs:

Instruction Type	Instruction Count (millions)	Cycles per Instruction
Machine A		
Arithmetic & logic	8	1
Load and store	4	3
Branch	2	4
Others	4	3
Machine B		
Arithmetic & logic	10	1
Load and store	8	2
Branch	2	4
Others	4	3

a. Determine the effective Cycles per Instruction (CPI), MIPS rate, and execution time for each machine. [3 Marks]

b. Comment on the results. [2 Marks]

safe pop
compare #1500, SP
Branch >= 0 ERROR
MOVE (SP)+, item

5. i.) There are two 8 bit registers, "R1" and "R2" contain the values -16 and 20 respectively. Find out the values of the status bits of V(overflow), N (Negative), Z(Zero), C(Carry) flags when the following instruction is executed.

SUB R2, R1 (Result in second operand)

[2 Marks]

ii.) Write an assembly routine program to ensure the stack (stack range from memory address 1100 to 1500), does not overflow or underflow before pushing or popping onto/from the stack. [3 Marks]

*** Best of Luck ***

$$R1 = R1 - R2$$

$$= R1 + (-R2)$$

Safe push compare #1500, SP
Branch <= 0 ERROR
MOVE ITEM, -(SP)

N = 0, S = 1 Z = 0, C = 1

A
CPI = 2.22
MIPS = 90
ET = 0.2 sec

B
CPI = 1.92
MIPS = 104
ET = 0.23 sec

11100000
-16 = 11110000
-20 = 11101100
-36 = 11011100